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Electrochemical device and electronic device

Abstract

An electrochemical device includes a housing, an electrode assembly and an electrolytic solution, and the electrode assembly and the electrolytic solution are disposed in the housing. The electrode assembly includes electrode plates and a separator disposed between adjacent electrode plates. The separator includes a porous substrate and a first bonding layer disposed on a surface of the porous substrate. The first bonding layer includes a plurality of first bonding regions spaced apart from each other. A thickness direction of the electrode assembly is defined as a first direction. In the first direction, a circumscribed rectangle of a projection of the electrode assembly possesses a diagonal line and a first side. An angle between the diagonal line and the first side is $\theta_{\text{sub.1}}$. An angle between a projection of each of the plurality of first bonding regions and the first side is $\theta_{\text{sub.2}}$, where $\theta_{\text{sub.1}}/2 \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}}/2 + 45^\circ)$.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of PCT international application Ser. No. PCT/CN2021/122859, filed on Oct. 9, 2021, which claims the benefit of priority to the Chinese Patent Application Ser. No. 202110338853.6, filed on Mar. 30, 2021, the disclosures of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) This application relates to the technical field of energy storage, and in particular, to an electrochemical device and an electronic device containing the electrochemical device.

BACKGROUND

(2) Electrochemical devices (such as a battery) are widely used in electronic devices such as an electronic mobile device, an electric tool, and an electric vehicle. With the development of such electronic devices, people pose higher requirements on the cycle performance of the electrochemical devices.

SUMMARY

(3) To overcome the foregoing disadvantages, it is necessary to provide an electrochemical device.

(4) In addition, it is necessary to provide an electronic device containing the electrochemical device.

(5) This application provides an electrochemical device, including a housing, an electrode assembly and an electrolytic solution, the electrode assembly and the electrolytic solution are disposed in the housing. The electrode assembly includes electrode plates and a separator disposed between adjacent electrode plates. The separator includes a porous substrate and a first bonding layer disposed on a surface of the porous substrate. The first bonding layer includes a plurality of first bonding regions spaced apart from each other. A thickness direction of the electrode assembly is defined as a first direction. In the first direction, a circumscribed rectangle of a projection of the electrode assembly possesses a diagonal line and a first side. An angle between the diagonal line and the first side is $\theta_{\text{sub.1}}$. An angle between a projection of each of the plurality of first bonding regions and the first side is $\theta_{\text{sub.2}}$, where $\theta_{\text{sub.1}/2} \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}/2} + 45^\circ)$.

(6) The first bonding layer disposed on the separator according to this application includes a plurality of spaced first bonding regions to improve the interfacial adhesion between the separator and the electrode plate. In addition, a gap between two adjacent first bonding regions reserves a space for transmitting the electrolytic solution, thereby meeting the requirement of achieving a trade-off between the interfacial adhesion and the electrolyte transmission. Further, it is set that $\theta_{\text{sub.1}/2} \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}/2} + 45^\circ)$ in this application, so that the first bonding regions tilt against the first side of the circumscribed rectangle of the projection of the electrode assembly in the thickness direction. Therefore, the bonding force of the first bonding layer is continuously distributed in the direction of the first side, thereby increasing the deformation resistance of the

electrochemical device. In addition, an appropriate relationship is set between $\theta_{\text{sub.2}}$ and $\theta_{\text{sub.1}}$, and therefore, a tilt angle of the first bonding regions deviates from a tilt angle of the diagonal line of the circumscribed rectangle by an amount that falls within a specified range. In this way, each first bonding region can form a relatively stable triangular fixing relationship with an end portion and a lateral portion of the electrode assembly, thereby further increasing the deformation resistance and cycle performance of the electrochemical device.

(7) In some possible implementations, the separator further includes a second bonding layer. The porous substrate is located between the first bonding layer and the second bonding layer. The second bonding layer includes a plurality of second bonding regions spaced apart from each other. Therefore, a gap between two adjacent second bonding regions can also be used as a channel for transmitting the electrolytic solution, thereby further improving the performance of transmitting the electrolytic solution. In the first direction, an angle between a projection of each of the plurality of second bonding regions and the projection of each of the plurality of first bonding regions is $\theta_{\text{sub.3}}$, where $0^\circ < \theta_{\text{sub.3}} < 180^\circ$. Therefore, the second bonding regions are also tilted, so that the bonding force of the second bonding layer is also continuously distributed in the direction of the first side, thereby further increasing the deformation resistance of the electrochemical device.

(8) In some possible implementations, $\theta_{\text{sub.3}} = 180^\circ - 2\theta_{\text{sub.2}}$. With the relationship between $\theta_{\text{sub.3}}$ and $\theta_{\text{sub.2}}$ further defined in this way, the deformation resistance of the electrode assembly is further increase.

(9) In some possible implementations, $\theta_{\text{sub.2}} = \theta_{\text{sub.1}}$. It is further defined that $\theta_{\text{sub.2}}$ is equal to $\theta_{\text{sub.1}}$, so that each first bonding region can form the stablest triangular fixing relationship with a second end portion and the lateral portion of the electrode assembly, thereby further increasing the deformation resistance of the electrode assembly.

(10) In some possible implementations, a direction from one of the first bonding regions to another adjacent first bonding region is defined as a second direction. In the second direction, a width of each of the plurality of first bonding regions is $d_{\text{sub.1}}$ mm, a distance between adjacent two of the plurality of first bonding regions is $d_{\text{sub.2}}$ mm, and $d_{\text{sub.1}}$ and $d_{\text{sub.2}}$ satisfy: $0.1 < d_{\text{sub.1}} < 10$, and $0.1 < d_{\text{sub.2}} < 10$. When $d_{\text{sub.1}}$ is exceptionally small, the bonding force between the first bonding layer and the electrode plate is reduced, and the deformation resistance is reduced accordingly. When $d_{\text{sub.1}}$ is exceptionally large, $d_{\text{sub.2}}$ is reduced accordingly, and the space available for transmitting the electrolytic solution is reduced, thereby deteriorating infiltration of the electrolytic solution. When $d_{\text{sub.2}}$ is exceptionally small, the space available for transmitting the electrolytic solution is also reduced, thereby deteriorating infiltration of the electrolytic solution. When $d_{\text{sub.2}}$ is exceptionally large, $d_{\text{sub.1}}$ is reduced accordingly, thereby also reducing the bonding force between the first bonding layer and the electrode plate and reducing the deformation resistance accordingly.

(11) In some possible implementations, a direction from one of the first bonding regions to another adjacent first bonding region is defined as a second direction. In the second direction, a width of the first bonding region is $d_{\text{sub.1}}$ mm, a distance between the adjacent first bonding regions is $d_{\text{sub.2}}$ mm, and $d_{\text{sub.1}}$ and $d_{\text{sub.2}}$ satisfy: $0.1 < d_{\text{sub.1}}/d_{\text{sub.2}} < 5$. When $d_{\text{sub.1}}/d_{\text{sub.2}}$ is exceptionally small, the proportion of the area of the first bonding regions is reduced, thereby also reducing the bonding force between the first bonding layer and the electrode plate, and reducing the deformation resistance accordingly. When $d_{\text{sub.1}}/d_{\text{sub.2}}$ is exceptionally large, the proportion of the area of the first bonding regions is increased, thereby also reducing the space available for transmitting the electrolytic solution and deteriorating infiltration of the electrolytic solution.

(12) In some possible implementations, in the first direction, a thickness of the first bonding layer is $t_{\text{sub.1}}$ μm , and $0.15 < t_{\text{sub.1}}d_{\text{sub.1}}/(d_{\text{sub.1}} + d_{\text{sub.2}}) < 2$. When $t_{\text{sub.1}}d_{\text{sub.1}}/(d_{\text{sub.1}} + d_{\text{sub.2}})$ is exceptionally small, the proportion of a diffusion area of the first bonding layer softened by heating is reduced, thereby also reducing the bonding force between the first bonding layer and the electrode plate, and reducing the deformation resistance accordingly. When

- t.sub.1d.sub.1/(d.sub.1+d.sub.2) is exceptionally large, the proportion of the diffusion area of the first bonding layer softened by heating is increased, thereby also reducing the space available for transmitting the electrolytic solution and deteriorating infiltration of the electrolytic solution.
- (13) In some possible implementations, in the first direction, the thickness of the first bonding layer is $t_{\text{sub.1}}$ μm , and $0.1 \leq t_{\text{sub.1}} \leq 6$. When $t_{\text{sub.1}}$ is exceptionally small, the bonding force between the first bonding layer and the electrode plate is reduced. When $t_{\text{sub.1}}$ is exceptionally large, thereby reducing overall air permeability of the separator and resulting in poor transmission of the electrolytic solution.
- (14) In some possible implementations, the plurality of the first bonding regions are parallel to each other. When the plurality of first bonding regions are parallel to each other, an angle may exist between two adjacent first bonding regions, where the angle is -5° to $+5^\circ$.
- (15) In some possible implementations, each of the plurality of first bonding regions is continuous or discontinuous. Gaps between different sections of each discontinuous first bonding region may also be used as a channel for transmitting the electrolytic solution, thereby further improving the performance of transmitting the electrolytic solution.
- (16) In some possible implementations, in the first direction, shapes of the projection of the electrode assembly include a rectangle, an arc, or an L shape. Therefore, the electrochemical device can fit with the shape of the internal space of the electronic device, so as to make full use of the space in the electronic device and increase the capacity of the electrochemical device.
- (17) In some possible implementations, the electrode assembly further includes tabs electrically connected to the electrode plates. Projections of the tabs in the first direction connect to an opposite side of the first side, so that the tabs can be connected to an external component conveniently.
- (18) In some possible implementations, structures of the electrode assembly include a wound structure or a stacked structure.
- (19) In some possible implementations, the porous substrate includes a polymer film, a multilayer polymer film, or a non-woven fabric, which, in each case, is formed by at least one of polyolefin, polyvinylidene difluoride, polyethylene terephthalate, cellulose, polyimide, polyamide, spandex, or polyphthalamide. Such polymers are of high thermal stability and make it easy to coat the porous substrate with the first bonding layer. In addition, the polymers are flexible and easily bendable.
- (20) In some possible implementations, the first bonding layer includes a bonding material. The bonding material includes at least one of poly(vinylidene difluoride-co-hexafluoropropylene), poly(vinylidene difluoride-co-trichloroethylene), polymethyl methacrylate, polyacrylic acid, polyacrylic acid sodium salt, polyacrylonitrile, polyvinylpyrrolidone, polyvinyl acetate, poly(ethylene-co-vinyl acetate), polyimide, polyethylene oxide, cellulose acetate, cellulose acetate butyrate, cellulose acetate propionate, cyanoethyl pullulan, cyanoethyl polyvinyl alcohol, cyanoethyl cellulose, cyanoethyl sucrose, pullulan, sodium carboxymethyl cellulose, lithium carboxymethyl cellulose, poly(acrylonitrile-co-styrene-co-butadiene), polyvinyl alcohol, polyvinyl ether, polytetrafluoroethylene, polyhexafluoropropylene, poly(styrene-co-butadiene), or polyvinylidene difluoride. Such polymers exert a strong bonding effect to bond the separator and the electrode plate together.
- (21) This application further provides an electronic device, including the electrochemical device described above.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is an overall schematic structural diagram of an electrochemical device according to an embodiment of this application;
- (2) FIG. 2 is a front view of an electrode assembly of the electrochemical device shown in FIG. 1

according to some embodiments;

(3) FIG. 3 is a front view of an electrode assembly of the electrochemical device shown in FIG. 1 according to some other embodiments;

(4) FIG. 4 is a front view of an electrode assembly of the electrochemical device shown in FIG. 1 according to still other embodiments;

(5) FIG. 5 is a top view of the electrode assembly shown in FIG. 2 according to some embodiments;

(6) FIG. 6 is a top view of the electrode assembly shown in FIG. 2 according to some other embodiments;

(7) FIG. 7 is a sectional view of the electrode assembly shown in FIG. 5 and sectioned along a VII-VII line;

(8) FIG. 8 is a front view of a separator of the electrode assembly shown in FIG. 2 according to some embodiments;

(9) FIG. 9 is a front view of a separator of the electrode assembly shown in FIG. 2 according to some other embodiments;

(10) FIG. 10 is a rear view of a separator of the electrode assembly shown in FIG. 2; and

(11) FIG. 11 is a schematic structural diagram of an electronic device according to an embodiment of this application.

REFERENCE NUMERALS OF MAIN COMPONENTS

(12) Electronic device **1** Housing **10** Body portion **11** Flank **12** Electrode assembly **20** First electrode plate **21** Second electrode plate **22** Separator **23** First tab **30** Second tab **40**

Electrochemical device **100** Top surface **111** Bottom surface **112** Upper surface **113** First lateral surface **114** Second lateral surface **115** Projection of the electrode assembly **200** First side **201** Second side **202** Diagonal line **203** First current collector **211** First active material layer **212** Second current collector **221** Second active material layer **222** Porous substrate **230** First bonding layer **231** Second bonding layer **232** First bonding region **2311** Second bonding region **2321** Circumscribed rectangle **R** Gaps **G.sub.1**, **G.sub.2**, **G.sub.3** First direction **D.sub.1** Second direction **D.sub.2** Third direction **D.sub.3** Fourth direction **D.sub.4** Width **d.sub.1** Distance **d.sub.2** Thickness **t.sub.1**

(13) This application is further described below with reference to the following specific embodiments and the foregoing drawings.

DETAILED DESCRIPTION

(14) The following describes the technical solutions in the embodiments of this application clearly and thoroughly with reference to the drawings herein. Evidently, the described embodiments are merely a part of but not all of the embodiments of this application.

(15) Unless otherwise defined, all technical and scientific terms used herein have the same meanings as usually understood by a person skilled in the technical field of this application. The terms used in the specification of this application herein are merely intended for describing specific embodiments but are not intended to limit this application.

(16) The following describes some embodiments of this application in detail with reference to drawings. To the extent that no conflict occurs, the following embodiments and the features in the embodiments may be combined with each other.

(17) An electrochemical device generally includes electrode plates and a separator disposed between adjacent electrode plates. Generally, a separator of high bonding performance is used so that the adjacent electrode plates and the separator are bonded firmly together to ensure high interfacial adhesion between the separator and the electrode plates, and to suppress expansion deformation caused by gassing inside the electrochemical device. However, the separator of high bonding performance makes interstices smaller between the separator and the electrode plate, and makes the electrode plate hardly permeable by the electrolytic solution. Especially, for a wound battery, uneven distribution of the electrolytic solution is more prone to occur at a corner region of

the electrode assembly, thereby leading to purple-speck lithium plating during cycles and impairing the cycle performance of the electrochemical device.

(18) Referring to FIG. 1, an embodiment of this application provides an electrochemical device **100**, including a housing **10**, an electrode assembly **20** (shown in FIG. 5 and FIG. 6), an electrolytic solution, a first tab **30** and a second tab **40**. The electrode assembly **20** and the electrolytic solution are accommodated in the housing **10**. The electrochemical device **100** shown in FIG. 5 and FIG. 6 includes one electrode assembly **20**. However, understandably, in order to achieve a high-voltage output, the electrochemical device **100** may include a plurality of electrode assemblies **20** connected in series instead, without being limited in this application.

(19) As shown in FIG. 5 and FIG. 6, the electrode assembly **20** includes electrode plates and a separator **23** disposed between adjacent electrode plates. The electrode plates include a first electrode plate **21** and a second electrode plate **22**. The separator **23** is disposed between the first electrode plate **21** and the second electrode plate **22**. The separator **23** is configured to prevent direct contact between the first electrode plate **21** and the second electrode plate **22**, where the direct contact may cause a short circuit. In some embodiments, the electrode assembly **20** is obtained by winding the first electrode plate **21**, the separator, and the second electrode plate **22**. In other words, the electrode assembly **20** is a wound structure. In some other embodiments, the electrode assembly **20** may be obtained by stacking the first electrode plate **21**, the separator, and the second electrode plate **22**. In other words, the electrode assembly **20** is a stacked structure.

(20) The first tab **30** and the second tab **40** are electrically connected to the electrode assembly **20**, respectively, and protrude from the housing **10** to connect to an external component (not shown in the drawing). Specifically, the first electrode plate **21** includes a first current collector **211** and a first active material layer **212** disposed on a surface of the first current collector **211**. The first tab **30** is electrically connected to the first current collector **211**. The second electrode plate **22** includes a second current collector **221** and a second active material layer **222** disposed on a surface of the second current collector **221**. The second tab **40** is electrically connected to the second current collector **221**. In some embodiments, the first electrode plate **21** is a positive electrode plate, and the second electrode plate **22** is a negative electrode plate.

(21) As shown in FIG. 1, the housing **10** may be a packaging bag obtained by sealing with a sealing film. In other words, the electrochemical device **100** may be a pouch-type battery. In this case, the housing **10** includes a body portion **11** and a sealing portion **12**. The body portion **11** is configured to accommodate the electrode assembly **20** and the electrolytic solution. The sealing portion **12** is connected to the body portion **11**. The first tab **30** and the second tab **40** protrude out of the sealing portion **12**. The body portion **11** includes a top surface **111**, a bottom surface **112**, an upper surface **113**, a lower surface (not shown), a first lateral surface **114**, and a second lateral surface **115**. The top surface **111** is opposite to the bottom surface **112**. The upper surface **113** is opposite to the lower surface, and is connected between the top surface **111** and the bottom surface **112**. The first lateral surface **114** is opposite to the second lateral surface **115**, and is connected between the top surface **111** and the bottom surface **112**. The sealing portion **12** is connected to the top surface **111**. In other embodiments, without being limited to a pouch-type battery, the electrochemical device **100** may be a steel- or aluminum-shell battery or the like instead. The type of the electrochemical device is not limited in this application.

(22) Referring to FIG. 7, the separator **23** includes a porous substrate **230** and a first bonding layer **231** disposed on a surface of the porous substrate **230**. The first bonding layer **231** bonds with the first electrode plate **21**. In another embodiment, the first bonding layer **231** may bond with the second electrode plate **22** instead.

(23) Referring to FIG. 8, the first bonding layer **231** includes a plurality of first bonding regions **2311** spaced apart from each other. In other words, adjacent two of the plurality of first bonding regions **2311** are interspaced with a gap G.sub.1. In some embodiments, the plurality of first bonding regions **2311** are parallel to each other. Understandably, when two components are referred

to as being parallel to each other, the two components are not necessarily parallel to each other exactly, but an angle may exist between the two components, where the angle is -5° to $+5^{\circ}$.

(24) It is defined that the thickness direction (that is, the direction from the upper surface **113** to the lower surface), the length direction (that is, the direction from the bottom surface **112** to the top surface **111**), and the width direction (that is, the direction from the first lateral surface **114** to the second lateral surface **115**) of the electrode assembly **20** are a first direction D.sub.1, a third direction D.sub.3, and a fourth direction D.sub.4, respectively. The third direction D.sub.3 is the direction in which the first tab **30** and the second tab **40** protrude. Referring to FIG. 2, in some embodiments, in the first direction D.sub.1, the projection **200** of the electrode assembly **20** is rectangular, and possesses a first side **201**, a second side **202**, and a diagonal line **203** connected to both the first side **201** and the second side **202**.

(25) As shown in FIG. 2, the electrode assembly **20** possesses a first end portion **20a**, a second end portion **20b**, and a lateral portion **20c** connected between the first end portion **20a** and the second end portion **20b**. The first end portion **20a** is a head of the electrode assembly **20**, and is a portion at which the first tab **30** and the second tab **40** are disposed. The second end portion **20b** is an ending of the electrode assembly **20**, where the ending is opposite to the head. The projection of the second end portion **20b** in the first direction D.sub.1 forms the first side **201**. The projection of the lateral portion **20c** in the first direction D.sub.1 forms the second side **202**. In this case, the projections of the first tab **30** and the second tab **40** in the first direction D.sub.1 connect to the opposite side of the first side **201**, so that the first tab **30** and the second tab **40** can be electrically connected to an external component conveniently.

(26) Referring to FIG. 2 and FIG. 8, it is defined that an angle between the diagonal line **203** and the first side **201** is $\theta_{\text{sub.1}}$. The diagonal line **203** of the circumscribed rectangle R tilts against the first side **201**, and therefore, $0^{\circ} < \theta_{\text{sub.1}} < 90^{\circ}$. The angle between the projection of each of the plurality of first bonding regions **2311** and the first side **201** is $\theta_{\text{sub.2}}$. $\theta_{\text{sub.1}}$ and $\theta_{\text{sub.2}}$ satisfy: $\theta_{\text{sub.1}}/2 \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}}/2 + 45^{\circ})$.

(27) As shown in FIG. 8, in some embodiments, each first bonding region **2311** may be strip-shaped and disposed on the surface of the porous substrate **230**. That is, each of the plurality of first bonding regions **2311** is continuous. In this case, the projection of each of the plurality of first bonding regions **2311** is also strip-shaped. The angle $\theta_{\text{sub.2}}$ between the projection of each of the plurality of first bonding regions **2311** and the first side **201** is the angle between the strip-shaped projection and the first side **201**. As shown in FIG. 9, in another embodiment, each of the plurality of first bonding regions **2311** may be discontinuous instead, for example, may be disposed on the surface of the porous substrate **230** in shapes of a plurality of blocks or islands. In this case, the projection of each of the plurality of first bonding regions **2311** is also in the shape of a plurality of blocks or islands. The angle $\theta_{\text{sub.2}}$ between the projection of each of the plurality of first bonding regions **2311** and the first side **201** is the angle between a connection line of the projections of the blocks or islands and the first side **201**.

(28) Gaps G.sub.2 also exist between sections of each discontinuous first bonding region **2311**, and can be used as a channel for transmitting the electrolytic solution, thereby further improving the performance of transmitting the electrolytic solution.

(29) The first bonding layer **231** disposed on the separator according to this application includes a plurality of spaced first bonding regions **2311**. The plurality of first bonding regions **2311** can bond with the first electrode plate **21** or the second electrode plate **22**, so as to improve the interfacial adhesion between the separator **23** and the electrode plate and suppress the expansion deformation caused by the gassing inside the electrochemical device **100**. This increases the deformation resistance of the electrochemical device **100**, reduces the hazards of deformation and structural disruption of the electrode assembly, and keeps smoothness of the entire conductive network in the electrode assembly **20**, thereby improving the cycle performance of the electrochemical device **100**. A gap G.sub.1 exists between adjacent two of the plurality of first bonding regions **2311** to reserve

a space for transmitting the electrolytic solution. After the electrolytic solution is injected, the electrolytic solution can flow through the gap G.sub.1 and infiltrate the electrode plate thoroughly, thereby also helping to improve the cycle performance of the electrochemical device **100**. The electrochemical device **100** according to this application achieves a trade-off between interfacial adhesion and the required capability of transmitting the electrolytic solution.

(30) Further, in this application, it is set that $\theta_{\text{sub.1}}/2 \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}}/2 + 45^\circ)$, $0^\circ < \theta_{\text{sub.1}} < 90^\circ$, and therefore, $0^\circ < \theta_2 < 90^\circ$, thereby causing the projections of the first bonding regions **2311** to also tilt against the first side **201**. Therefore, the bonding force of the first bonding layer **231** is continuously distributed in the direction of the first side **201**. That is, the bonding force between the separator **23** and the electrode plate is continuously distributed in the direction of the first side **201**, thereby increasing the deformation resistance of the electrochemical device **100** and improving the cycle performance. In addition, an appropriate relationship is set between $\theta_{\text{sub.2}}$ and $\theta_{\text{sub.1}}$, and therefore, a tilt angle of the first bonding regions **2311** deviates from a tilt angle of the diagonal line **203** of the circumscribed rectangle R by an amount that falls within a specified range. Therefore, each first bonding region **2311** can form a relatively stable triangular fixing relationship with the second end portion **20b** and the lateral portion **20c** of the electrode assembly **20**, thereby further increasing the deformation resistance of the electrochemical device **100** and improving the cycle performance.

(31) Understandably, in order to improve the competitiveness of electronic devices, some consumer electronic devices are changing toward irregular shapes. An electrochemical device in an irregular shape can make full use of the irregular space in such electronic devices, thereby increasing the capacity of the entire electrochemical device. Therefore, in this application, in the first direction D.sub.1, the shape of the projection **200** of the electrode assembly **20** is not limited to a rectangle, and may be other irregular shapes instead. As shown in FIG. 3 and FIG. 4, in another embodiment, when the electrode assembly **20** is a stacked structure, shapes of the projection **200** of the electrode assembly **20** in the first direction D.sub.1 include an arc or an L shape. In this case, the circumscribed rectangle R of the projection **200** of the electrode assembly **20** also possesses a first side **201**, a second side **202**, and a diagonal line **203** connected between the first side **201** and the second side **202**. Understandably, as shown in FIG. 2, when the projection **200** of the electrode assembly **20** is rectangular, the circumscribed rectangle R of the projection **200** is the projection **200** itself.

(32) In some embodiments, the porous substrate **230** includes a polymer film, a multilayer polymer film, or a non-woven fabric, which, in each case, is formed by at least one of polyolefin, polyvinylidene difluoride, polyethylene terephthalate, cellulose, polyimide, polyamide, spandex, or polyphthalamide. Such polymers possess high thermal stability, and facilitate surface treatment, thereby making it easy to coat the porous substrate **230** with the first bonding layer **231**. In addition, such polymers are highly flexible and bendable.

(33) The first bonding layer **231** includes a bonding material. The bonding material includes at least one of the following polymers: poly(vinylidene difluoride-co-hexafluoropropylene), poly(vinylidene difluoride-co-trichloroethylene), polymethyl methacrylate, polyacrylic acid, polyacrylic acid sodium salt, polyacrylonitrile, polyvinylpyrrolidone, polyvinyl acetate, poly(ethylene-co-vinyl acetate), polyimide, polyethylene oxide, cellulose acetate, cellulose acetate butyrate, cellulose acetate propionate, cyanoethyl pullulan, cyanoethyl polyvinyl alcohol, cyanoethyl cellulose, cyanoethyl sucrose, pullulan, sodium carboxymethyl cellulose, lithium carboxymethyl cellulose, poly(acrylonitrile-co-styrene-co-butadiene), polyvinyl alcohol, polyvinyl ether, polytetrafluoroethylene, polyhexafluoropropylene, poly(styrene-co-butadiene), or polyvinylidene difluoride. Such polymers exert a strong bonding effect to bond the separator **23** and the first electrode plate **21** or the second electrode plate **22** together.

(34) As shown in FIG. 7, in some embodiments, the separator **23** further includes a second bonding layer **232**. The porous substrate **230** is located between the first bonding layer **231** and the second

bonding layer **232**. When the first bonding layer **231** bonds with the first electrode plate **21**, the second bonding layer **232** bonds with the second electrode plate **22**. Referring to FIG. **10**, the second bonding layer **232** includes a plurality of second bonding regions **2321** spaced apart from each other. In other words, two adjacent second bonding regions **2321** are also interspaced with a gap $G_{sub.3}$ that serves as a channel for transmitting the electrolytic solution. Therefore, the second bonding regions **2321** disposed herein further improve the infiltration effect of the electrolytic solution. In some embodiments, a plurality of second bonding regions **2321** are parallel to each other.

(35) In the first direction $D_{sub.1}$, an angle between the projection **200** of the second bonding regions **2321** and the projection **200** of the first bonding regions **2311** is $\theta_{sub.3}$, where $0^\circ < \theta_{sub.3} < 180^\circ$. In other words, the second bonding regions **2321** also tilt against the first side **201**. Therefore, the bonding force of the second bonding layer **232** is also continuously distributed in the direction of the first side **201**, thereby further increasing the deformation resistance of the electrochemical device **100**.

(36) Further, $\theta_{sub.2}$ and $\theta_{sub.3}$ may further satisfy: $\theta_{sub.3} = 180^\circ - 2\theta_{sub.2}$. With the relationship between $\theta_{sub.3}$ and $\theta_{sub.2}$ further defined in this way, the deformation resistance of the electrode assembly **20** is further increased.

(37) In some embodiments, $\theta_{sub.2} = \theta_{sub.1}$. It is further defined that $\theta_{sub.2}$ is equal to $\theta_{sub.1}$, so that each first bonding region **2311** can form the stablest triangular fixing relationship with a second end portion **20b** and the lateral portion **20c** of the electrode assembly **20**, thereby further increasing the deformation resistance of the electrode assembly **20**.

(38) As shown in FIG. **8**, in some embodiments, a direction from one of the first bonding regions **2311** to another adjacent first bonding region **2311** is defined as a second direction $D_{sub.2}$. In the second direction $D_{sub.2}$, a width of the first bonding region **2311** is $d_{sub.1}$ mm, a distance between the adjacent first bonding regions **2311** is $d_{sub.2}$ mm, and $d_{sub.1}$ and $d_{sub.2}$ satisfy: $0.1 < d_{sub.1} < 10$, and $0.1 < d_{sub.2} < 10$. When $d_{sub.1}$ is exceptionally small, the bonding force between the first bonding layer **231** and the electrode plate is reduced, and the deformation resistance is reduced accordingly. When $d_{sub.1}$ is exceptionally large, $d_{sub.2}$ is reduced accordingly on condition that the area of the electrode plate is constant, and the space available for transmitting the electrolytic solution is reduced, thereby deteriorating infiltration of the electrolytic solution. When $d_{sub.2}$ is exceptionally small, the space available for transmitting the electrolytic solution is also reduced, thereby deteriorating infiltration of the electrolytic solution. When $d_{sub.2}$ is exceptionally large, $d_{sub.1}$ is reduced accordingly on condition that the area of the electrode plate is constant, thereby also reducing the bonding force between the first bonding layer **231** and the electrode plate and reducing the deformation resistance accordingly.

(39) In some embodiments, $d_{sub.1}$ and $d_{sub.2}$ satisfy: $0.1 < d_{sub.1}/d_{sub.2} < 5$. When $d_{sub.1}/d_{sub.2}$ is exceptionally small, the proportion of the area of the first bonding regions **2311** is reduced, thereby also reducing the bonding force between the first bonding layer **231** and the electrode plate, and reducing the deformation resistance accordingly. When $d_{sub.1}/d_{sub.2}$ is exceptionally large, the proportion of the area of the first bonding regions **2311** is increased, thereby also reducing the space available for transmitting the electrolytic solution and deteriorating infiltration of the electrolytic solution.

(40) In some embodiments, in the first direction $D_{sub.1}$, a thickness of the first bonding layer **231** is $t_{sub.1}$ μm , and $0.15 < t_{sub.1}d_{sub.1}/(d_{sub.1} + d_{sub.2}) < 2$. When $t_{sub.1}d_{sub.1}/(d_{sub.1} + d_{sub.2})$ is exceptionally small, the proportion of a diffusion area of the first bonding layer **231** softened by heating is reduced, thereby also reducing the bonding force between the first bonding layer **231** and the electrode plate, and reducing the deformation resistance accordingly. When $t_{sub.1}d_{sub.1}/(d_{sub.1} + d_{sub.2})$ is exceptionally large, the proportion of the diffusion area of the first bonding layer **231** softened by heating is increased, thereby also reducing the space available for transmitting the electrolytic solution and deteriorating infiltration of the electrolytic solution.

(41) Referring to FIG. 7 and FIG. 8, in some embodiments, $0.1 \leq t_{\text{sub.1}} \leq 6$. When $t_{\text{sub.1}}$ is exceptionally small, the bonding force between the first bonding layer 231 and the electrode plate is reduced. When $t_{\text{sub.1}}$ is exceptionally large, thereby reducing overall air permeability of the separator 23 and resulting in poor transmission of the electrolytic solution.

(42) In this application, the electrochemical device 100 may be any type of primary batteries, secondary batteries, fuel cells, solar cells, and capacitors (such as supercapacitors). Especially, the secondary battery may be a lithium secondary battery, including a lithium metal secondary battery, a lithium-ion secondary battery, a lithium polymer secondary battery, and a lithium-ion polymer secondary battery.

(43) Referring to FIG. 11, an embodiment of this application further provides an electronic device 1. The electronic device 1 includes the electrochemical device 100 disclosed above. The electronic device 1 according to this application may be, but is not limited to: a notebook computer, a pen-inputting computer, a mobile computer, an e-book player, a portable phone, a portable fax machine, a portable photocopier, a portable printer, a stereo headset, a video recorder, a liquid crystal display television set, a handheld cleaner, a portable CD player, a mini CD-ROM, a transceiver, an electronic notepad, a calculator, a memory card, a portable voice recorder, a radio, a backup power supply, a motor, a car, a motorcycle, a power-assisted bicycle, a bicycle, a lighting appliance, a toy, a game machine, a watch, an electric tool, a flashlight, a camera, a large household battery, a lithium-ion capacitor, and the like.

(44) The following describes this application in detail with reference to specific embodiments and comparative embodiments. This application is described below with reference to a specific preparation process and a test method using a pouch-type cell as an example of the electrochemical device. A person skilled in the art understands that the preparation method described in this application is merely an example. Any other appropriate preparation methods fall within the scope of this application.

Comparative Embodiment 1

(45) The separator is coated on both sides with a first bonding layer and a second bonding layer. The electrode assembly is obtained by sequentially stacking and winding the first electrode plate, the second electrode plate, and the separator. In the first direction, a projection of the electrode assembly is rectangular. An angle $\theta_{\text{sub.1}}$ between a diagonal line of the projection and the first side is 45° . An angle $\theta_{\text{sub.2}}$ between a projection of first bonding regions of the first bonding layer and the first side is 90° . An angle $\theta_{\text{sub.3}}$ between a projection of second bonding regions of the second bonding layer and the projection of the first bonding regions is 23° . A width $d_{\text{sub.1}}$ of each of the first bonding regions is 0.21 mm, a distance $d_{\text{sub.2}}$ between adjacent first bonding regions is 0.04 mm, and the thickness $t_{\text{sub.1}}$ of the first bonding region is 2 μm .

(46) The electrode assembly and the electrolytic solution are packaged into an aluminum plastic film to obtain an electrochemical device.

Comparative Embodiments 2 to 5 and Embodiments 1 to 12

(47) Differences from Comparative Embodiment 1 are: the value of $\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, or $\theta_{\text{sub.3}}$ is different; the second bonding layer is omitted in the separator; or, the first bonding layer and the second bonding layer are evenly applied as coating (that is, the entire surface of the porous substrate is evenly coated with the bonding layers).

(48) A capacity retention rate test and a three-point bending test are performed on the electrochemical device in each embodiment and each comparative embodiment, and the corresponding test results are recorded in Table 1. The steps of the capacity retention rate test include: charging the electrochemical device at a constant current of 1 C in a 45°C . muffle oven until the voltage reaches 4.43 V, then charging the electrochemical device at a constant voltage until the current reaches 0.05 C, then leaving the electrochemical device to stand for 5 minutes; and then discharging the electrochemical device at a current of 0.7 C until the voltage reaches 3.0 V, whereupon the discharge capacity is a first-cycle discharge capacity of the electrochemical device,

recorded as 100%. The foregoing charging and discharging steps are repeated for 400 cycles separately. A ratio of the discharge capacity of the electrochemical device at the end of the cycles to the first-cycle discharge capacity, multiplied by 100%, is the capacity retention rate.

(49) The steps of the three-point bending test include: fixing the electrochemical device onto two supporting points that are a distance apart from each other in a dry environment, and using an indenter to apply a pressure to the electrochemical device at a midpoint of the two supporting points; and then recording a displacement by which the electrochemical device is pressed downward with the indenter, and the pressure that is withstood by the electrochemical device and that corresponds to the displacement. When the recorded pressure value jumps (plunges), it indicates that the electrochemical device is fractured inside. The pressure value at this time may be used to represent the hardness of the electrochemical device.

(50) TABLE-US-00001 TABLE 1 400.sup.th-cycle Three-point capacity bending $\theta_{sub.1}$ (°) $\theta_{sub.2}$ (°) $\theta_{sub.3}$ (°) retention rate (%) pressure (N) Comparative 45 90 23 67 170 Embodiment 1 Comparative 45 90 30 65 167 Embodiment 2 Comparative 45 90 45 67 165 Embodiment 3 Comparative 45 90 90 68 180 Embodiment 4 Comparative 45 Coated Coated 70 190 Embodiment 5 evenly evenly Embodiment 1 45 45 23 81 181 Embodiment 2 45 45 30 83 183 Embodiment 3 45 45 86 188 Embodiment 4 45 45 90 92 192 Embodiment 5 45 45 120 87 190 Embodiment 6 45 45 160 80 183 Embodiment 7 45 20 140 81 185 Embodiment 8 45 50 80 83 190 Embodiment 9 45 75 30 82 184 Embodiment 62 45 90 78 181 10 Embodiment 62 62 90 88 190 11 Embodiment 62 62 56 94 196 12

(51) As can be seen from the data in Table 1, compared with Comparative Embodiments 1 to 4, the electrochemical devices in Embodiments 1 to 4 possess relatively high hardness (that is, high deformation resistance) and an improved capacity retention rate because Embodiments 1 to 4 satisfy $\theta_{sub.1}/2 \leq \theta_{sub.2} \leq (\theta_{sub.1}/2 + 45^\circ)$. The electrochemical device in Embodiment 4 achieves even a higher deformation resistance and a higher capacity retention rate due to $\theta_{sub.2} = \theta_{sub.1}$. The electrochemical device in Comparative Embodiment 5 achieves the lowest cycle performance because the separator is coated evenly with the bonding layer.

(52) Further, as can be seen from comparison between Embodiments 1 to 6, when $\theta_{sub.1}/2 \leq \theta_{sub.2} \leq (\theta_{sub.1}/2 + 45^\circ)$, as $\theta_{sub.3}$ increases, the deformation resistance and the capacity retention rate of the electrochemical device show a tendency to increase first and then decrease. When $\theta_{sub.3} = 180^\circ - 2\theta_{sub.2}$, the electrochemical device achieves a higher deformation resistance and a higher capacity retention rate. Therefore, the electrochemical device in Embodiment 4 possesses even a higher deformation resistance and a higher capacity retention rate.

(53) Still further, as can be seen from comparison between Embodiments 4 and 7 to 10, when $\theta_{sub.3} = 180^\circ - 2\theta_{sub.2}$, as $\theta_{sub.2}$ increases, the deformation resistance and the capacity retention rate of the electrochemical device show a tendency to increase first and then decrease. When $\theta_{sub.2} = \theta_{sub.1}$, the electrochemical device achieves even a higher deformation resistance and a higher capacity retention rate. Therefore, the electrochemical device in Embodiment 4 possesses even a higher deformation resistance and a higher capacity retention rate.

(54) Moreover, when $\theta_{sub.2} = \theta_{sub.1}$, the specific value of $\theta_{sub.1}$ is adjustable. In contrast with Embodiment 10, the electrochemical devices in Embodiments 11 to 12 achieve even a higher deformation resistance and a higher capacity retention rate due to $\theta_{sub.2} = \theta_{sub.1}$. Embodiments 13 to 26

(55) Differences from Embodiment 1 are that the values of $t_{sub.1}$, $d_{sub.1}$, and $d_{sub.2}$ are further set, and the value of $t_{sub.1}$, $d_{sub.1}$, or $d_{sub.2}$ varies in Embodiments 13 to 26.

(56) A capacity retention rate test and a three-point bending test are performed on the electrochemical device in each embodiment and each comparative embodiment, and the corresponding test results are recorded in Table 2.

(57) TABLE-US-00002 TABLE 2 400.sup.th-cycle Three-point $d_{sub.1}$ $d_{sub.2}$ $t_{sub.1}$ capacity bending (mm) (mm) (μm) $d_{sub.1}/d_{sub.2}$ $t_{sub.1}d_{sub.1}/(d_{sub.1} + d_{sub.2})$ retention rate pressure

(N) Embodiment 13 0.21 0.04 2 5.25 1.68 70.2 193 Embodiment 14 0.21 0.06 2 3.5 1.56 80.4 189
 Embodiment 15 0.21 0.12 2 1.79 1.28 88.4 185 Embodiment 16 0.21 0.31 2 0.679 0.81 85.5 170
 Embodiment 17 0.21 2.2 2 0.095 0.17 84 150 Embodiment 18 0.21 12 2 0.018 0.03 78.7 130
 Embodiment 19 0.52 0.31 2 1.682 1.25 87.4 185 Embodiment 20 0.83 0.31 2 2.684 1.46 90.4 188
 Embodiment 21 4.02 0.31 2 13.002 1.86 70.2 192 Embodiment 22 12 12 2 1 1 80.1 167
 Embodiment 23 0.52 0.31 0.5 1.682 0.31 80.2 163 Embodiment 24 0.52 0.31 2.4 1.682 1.51 88.3
 181 Embodiment 25 0.52 0.31 5.6 1.682 3.51 75.9 191 Embodiment 26 0.52 0.31 8.2 1.682 5.14
 69.9 194

(58) Under the condition that $\theta_{\text{sub.1}/2} \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}/2} + 45^\circ)$, the value of $d_{\text{sub.2}}$ also affects the capacity retention rate of the electrochemical device. As can be seen from the data in Table 2, in contrast with Embodiments 13 to 17, the bonding force between the separator and the electrode plate in Embodiment 18 is relatively reduced due to $d_{\text{sub.2}}$ greater than 10 mm, thereby relatively reducing the hardness of the electrochemical device (that is, deformation resistance), and also relatively reducing the capacity retention rate.

(59) Secondly, the value of $d_{\text{sub.1}}$ also affects the capacity retention rate of the electrochemical device. In contrast with Embodiments 16 and 19 to 21, the space available for transmitting the electrolytic solution and the capacity retention rate in Embodiment 22 are relatively reduced due to $d_{\text{sub.1}}$ greater than 10 mm.

(60) Thirdly, the value of $d_{\text{sub.1}}/d_{\text{sub.2}}$ also affects the capacity retention rate of the electrochemical device. In contrast with Embodiments 19, 20, and 22, the capacity retention rate of the electrochemical device in Embodiment 21 is relatively reduced due to $d_{\text{sub.1}}/d_{\text{sub.2}}$ greater than 5.

(61) Finally, the value of $t_{\text{sub.1}}d_{\text{sub.1}}/(d_{\text{sub.1}}+d_{\text{sub.2}})$ also affects the capacity retention rate of the electrochemical device. In contrast with Embodiments 19 and 23 to 24, the capacity retention rate of the electrochemical device in Embodiments 25 to 26 is relatively reduced due to $t_{\text{sub.1}}d_{\text{sub.1}}/(d_{\text{sub.1}}+d_{\text{sub.2}})$ greater than 2.

(62) The foregoing embodiments are merely intended for describing the technical solutions of this application but not intended as a limitation. Although this application is described in detail with reference to the foregoing optional embodiments, a person of ordinary skill in the art understands that modifications or equivalent replacements may be made to the technical solutions of this application.

Claims

1. An electrochemical device, comprising: a housing, an electrode assembly and an electrolytic solution; the electrode assembly and the electrolytic solution are disposed in the housing; and the electrode assembly comprises electrode plates and a separator disposed between adjacent electrode plates; wherein the separator comprises a porous substrate, a first bonding layer, and a second bonding layer, the porous substrate is located between the first bonding layer and the second bonding layer; the first bonding layer comprises a plurality of first bonding regions spaced apart from each other, and the second bonding layer comprises a plurality of second bonding regions spaced apart from each other; a thickness direction of the electrode assembly is defined as a first direction; in the first direction, a circumscribed rectangle of a projection of the electrode assembly possesses a diagonal line and a first side, an angle between the diagonal line and the first side is $\theta_{\text{sub.1}}$, and an angle between a projection of each of the plurality of first bonding regions and the first side is $\theta_{\text{sub.2}}$, wherein $\theta_{\text{sub.1}}/2 \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}}/2 + 45^\circ)$; in the first direction, an angle between a projection of each of the plurality of second bonding regions and the projection of each of the plurality of first bonding regions is $\theta_{\text{sub.3}}$, wherein $45^\circ < \theta_{\text{sub.3}} \leq 120^\circ$.
2. The electrochemical device according to claim 1, $80^\circ \leq \theta_{\text{sub.3}} \leq 120^\circ$.
3. The electrochemical device according to claim 1, wherein $\theta_{\text{sub.3}} = 180^\circ - 2\theta_{\text{sub.2}}$.

4. The electrochemical device according to claim 1, wherein $\theta_{\text{sub.2}} = \theta_{\text{sub.1}}$.
5. The electrochemical device according to claim 1, wherein a direction from one of the plurality of first bonding regions to another adjacent one of the plurality of first bonding regions is defined as a second direction, and in the second direction, a width of each of the plurality of first bonding regions is $d_{\text{sub.1}}$ mm, a distance between adjacent two of the plurality of first bonding regions is $d_{\text{sub.2}}$ mm, and $d_{\text{sub.1}}$ and $d_{\text{sub.2}}$ satisfy at least one of the following conditions:
 $0.1 < d_{\text{sub.1}} < 10$, and $0.1 < d_{\text{sub.2}} < 10$; and
 $0.1 < d_{\text{sub.1}}/d_{\text{sub.2}} < 5$.
6. The electrochemical device according to claim 5, wherein in the first direction, a thickness of the first bonding layer is $t_{\text{sub.1}}$ μm , and $0.15 < t_{\text{sub.1}}d_{\text{sub.1}}/(d_{\text{sub.1}} + d_{\text{sub.2}}) < 2$.
7. The electrochemical device according to claim 1, wherein in the first direction, a thickness of the first bonding layer is $t_{\text{sub.1}}$ μm , and $0.1 \leq t_{\text{sub.1}} \leq 6$.
8. The electrochemical device according to claim 1, wherein the plurality of the first bonding regions are parallel to each other.
9. The electrochemical device according to claim 1, wherein each of the plurality of first bonding regions is discontinuous.
10. The electrochemical device according to claim 1, wherein in the first direction, shapes of the projection of the electrode assembly comprise a rectangle, an arc, or an L shape.
11. The electrochemical device according to claim 1, wherein the electrode assembly further comprises tabs electrically connected to the electrode plates, and projections of the tabs in the first direction connect to an opposite side of the first side.
12. The electrochemical device according to claim 1, wherein structures of the electrode assembly comprise a wound structure or a stacked structure.
13. The electrochemical device according to claim 1, wherein the porous substrate comprises a polymer film, a multilayer polymer film, or a non-woven fabric formed by at least one of polyolefin, polyvinylidene difluoride, polyethylene terephthalate, cellulose, polyimide, polyamide, spandex, or polyphthalamide.
14. The electrochemical device according to claim 1, wherein the first bonding layer comprises a bonding material, and the bonding material comprises at least one of poly(vinylidene difluoride-co-hexafluoropropylene), poly(vinylidene difluoride-co-trichloroethylene), polymethyl methacrylate, polyacrylic acid, polyacrylic acid sodium salt, polyacrylonitrile, polyvinylpyrrolidone, polyvinyl acetate, poly(ethylene-co-vinyl acetate), polyimide, polyethylene oxide, cellulose acetate, cellulose acetate butyrate, cellulose acetate propionate, cyanoethyl pullulan, cyanoethyl polyvinyl alcohol, cyanoethyl cellulose, cyanoethyl sucrose, pullulan, sodium carboxymethyl cellulose, lithium carboxymethyl cellulose, poly(acrylonitrile-co-styrene-co-butadiene), polyvinyl alcohol, polyvinyl ether, polytetrafluoroethylene, polyhexafluoropropylene, poly(styrene-co-butadiene), or polyvinylidene difluoride.
15. An electronic device, comprising an electrochemical device, the electrochemical device comprises a housing, an electrode assembly and an electrolytic solution; the assembly and the electrolytic are disposed in the housing; and the electrode assembly comprises electrode plates and a separator disposed between adjacent electrode plates; wherein the separator comprises a porous substrate, a first bonding layer, and a second bonding layer, the porous substrate is located between the first bonding layer and the second bonding layer, the first bonding layer comprises a plurality of first bonding regions spaced apart from each other, and the second bonding layer comprises a plurality of second bonding regions spaced apart from each other; a thickness direction of the electrode assembly is defined as a first direction; in the first direction, a circumscribed rectangle of a projection of the electrode assembly possesses a diagonal line and a first side, an angle between the diagonal line and the first side is $\theta_{\text{sub.1}}$, and an angle between a projection of each of the plurality of first bonding regions and the first side is $\theta_{\text{sub.2}}$, wherein $\theta_{\text{sub.1}}/2 \leq \theta_{\text{sub.2}} \leq (\theta_{\text{sub.1}}/2 +) 45^\circ$; in the first direction, an angle between a projection of each of

the plurality of second bonding regions and the projection of each of the plurality of first bonding regions is $\theta_{\text{sub.3}}$, wherein $45^\circ \leq \theta_{\text{sub.3}} \leq 120^\circ$.

16. The electronic device according to claim 15, wherein $80^\circ < \theta_{\text{sub.3}} < 120^\circ$.

17. The electronic device according to claim 15, wherein $\theta_{\text{sub.3}} = 180^\circ - 2\theta_{\text{sub.2}}$.

18. The electronic device according to claim 15, wherein a direction from one of the plurality of first bonding regions to another adjacent one of the plurality of first bonding region is defined as a second direction, and in the second direction, a width of each of the plurality of first bonding regions is $d_{\text{sub.1}}$ mm, a distance between adjacent two of the plurality of first bonding regions is $d_{\text{sub.2}}$ mm, and $d_{\text{sub.1}}$ and $d_{\text{sub.2}}$ satisfy at least one of the following conditions:

$0.1 < d_{\text{sub.1}} < 10$, and $0.1 < d_{\text{sub.2}} < 10$; and

$0.1 < d_{\text{sub.1}}/d_{\text{sub.2}} < 5$.

19. The electronic device according to claim 18, wherein in the first direction, a thickness of the first bonding layer is $t_{\text{sub.1}}$ μm , and $0.15 < t_{\text{sub.1}}d_{\text{sub.1}}/(d_{\text{sub.1}} + d_{\text{sub.2}}) < 2$.

20. The electronic device according to claim 15, wherein in the first direction, a thickness of the first bonding layer is $t_{\text{sub.1}}$ μm , and $0.1 \leq t_{\text{sub.1}} \leq 6$.
